technology stands on a strong foundation of more than 40 years of scientific research in cryptography, hashing, peer-to-peer networks, and consensus protocols. Figure 1.1 provides a brief history of blockchain, its innovation and robust scientific foundation, and its transformative effect on modern networked systems.

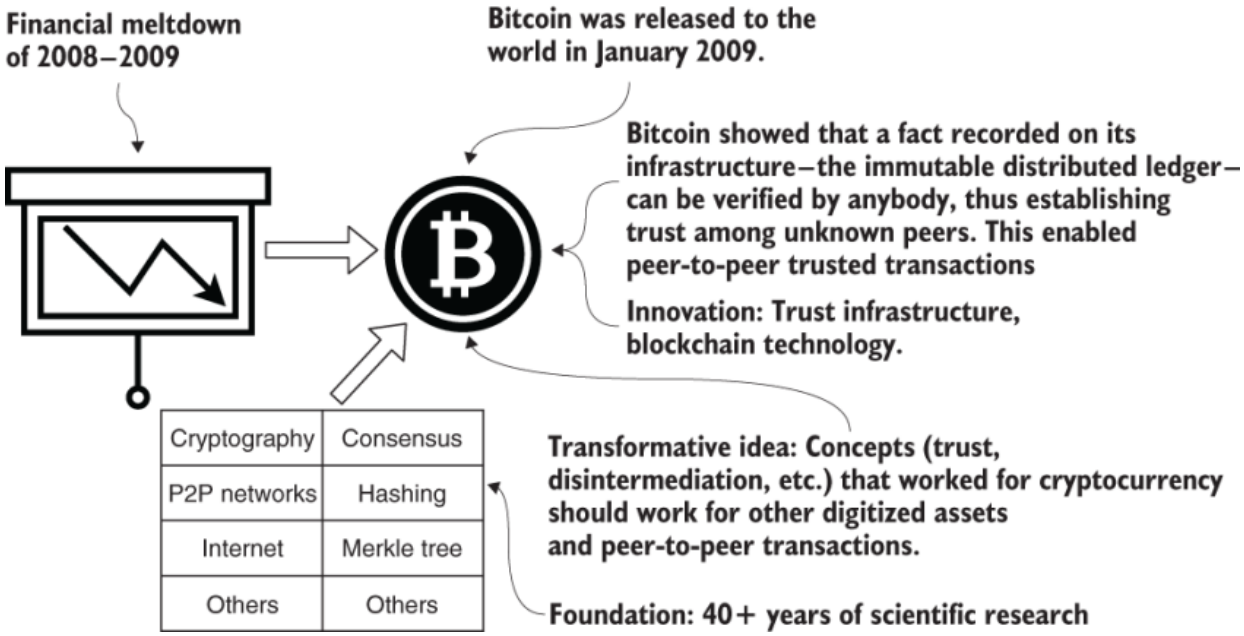


Figure 1.1 The birth of blockchain technology

On completion of this chapter, you'll know the fundamental concepts of blockchain and decentralized applications, such as transactions, blocks, a chain of blocks, nodes, networks of nodes, and the protocol that ties all these elements together. With these many components, blockchain is indeed a complex system. An understanding of these foundational concepts is, therefore, imperative for the blockchain application design and development that you'll embark on in chapters 2-11.